# Refining the OPALS-AWF Impact Project

10th September 2025

**Update on the collaborative project between the Oppenheimer Programme in African Landscape Systems (OPALS) and the African Wildlife Foundation (AWF), to optimise Isaac Mureithi's forthcoming 12-month placement with AWF.**

## Background

This impact-oriented applied research project is intended to support better models of landscape-scale Conservation Governance, through a collaborative training-position created between the AWF and the OPALS Programme (https://opals-exeter.org/) led by the University of Exeter. Oppenheimer Impact Scholar Isaac Nduta has a background in actuarial science and has just finished his 12-month MSc in Conservation and Biodiversity where he developed skills into applied geospatial ecology. His dissertation project titled "*Using the Relative Productivity Index to assess how governance structures and local community engagement influence rangeland health outcomes*" applied the Relative Productivity Index (RPI), a new indicator of rangeland condition (Lomax et al. 2025), to undertake a novel evaluation of spatial and temporal patterns that relate to local governance/management influences (see Appendix for details).

Our aim for this Impact project is for Isaac to make a material contribution to AWFs operations to support ecosystem-level conservation in the Amboseli & Tsavo landscapes, going beyond species-level protection to integrate the health and connectivity of the entire ecosystem. In the first phase of his post, Isaac has been managed by Dr Andy Cunliffe, in partnership with Dr Philip Muruthi, Dr Tom Powell, and Dr Kimberley Hockings. From the 1st of October, Isaac will commence a 12-month placement at AWF's Field Office in the Tsavo Landscape, co-managed by Kenneth Kimitei and Andy Cunliffe. Note that we have a commitment to open research (reports, data, models, code that Isaac works on will be made available openly), and Isaac will present on this project at the Oppenheimer Research Conference in South Africa in October 2026.

## Proposed focus for project Phase 2

The proposed focus of Isaac's contribution in Phase 2 of our partnership is to **explore the management applications of the Relative Productivity Index (RPI). This would include clarifying the value of RPI as a tool for adaptive management in rangeland landscapes and how RPI could be integrated into AWFs landscape management workflows both as a monitoring and evaluation indicator and to identify degradation hotspots for prioritisation in restoration areas.** This objective should aid landscape level conservation and impact-linked funding allocation. This focus has been endorsed by David Williams (AWF's geospatial lead) and received provisional endorsement from , it now requires consideration and feedback by more stakeholders from AWF.

If we proceed with this priority, Isaac will need to focus on delivery of this objective for perhaps 80% of his time over the coming year, as he'll need to continue his technical training to undertake advanced geospatial modelling using machine learning to first produce, and then evaluate, RPI maps at a finer (30 m) spatial grain to improve correspondence with management-relevant spatial scales. This work would utilise a recently released GPP product (https://landcarbonlab.org/insights/global-productivity-data-ecosystem-health and https://peerj.com/articles/19774/) and the University of Exeter's computing facilities. The RPI illuminates the recent past, currently with a ca. 12-18 month latency, so while it can reveal areas of concern/instabilities before they might be apparent in other indicators is not currently a very 'early' warning system. The RPI will end up as one of a suite of metrics that can be used in a complimentary way by land managers with suitable context to understand landscape condition.

Supporting this primary objective, secondary objectives should include:

1) **Communicate dissertation research findings to wider audiences** by (i) writing a short and accessible 3-page summary of dissertation findings for wider dissemination and (ii) refining dissertation findings for publication as a scientific paper.
2) **Gather high-quality spatial data on AWF interventions** by working with stakeholders to collate existing and collect new geospatial records of where, when, and what interventions have been undertaken by AWF through desk-based and field-based digitisation and interviews.

Tertiary objectives if there is spare capacity could include:

3) **Support AWFs policy and advocacy efforts to advance landscape-level conservation.** Isaac could support any aspect of AWFs policy and advocacy efforts, including but not limited to utilising his actuarial background to support innovations in financing landscape-scale conservation and/or supporting AWFs engagement with larger scale initiatives such as implementing Kenya's National Biodiversity Strategy and Action Plan (NBSAP).
4) **Support uptake of Open Research tools and approaches.** To increase organisational awareness of options to enhance transparency and integrity in workflows, and potentially lower expenditure on software licence fees, Isaac could run a training session for AWF colleagues on Open Research practice (https://tess-lab.org/resources/open-research/) and Free and Open Source Software tools (https://tess-lab.org/resources/free-and-open-source-software-used-in-tess-lab/).

## Provisional Specific Steps if adopting the primary objective proposed above:

- **Implement RPI modelling** using the machine learning in R (MLR3) ecosystem on TESS Lab high-performance workstations at the University of Exeter. Upgrade the RPI machine learning algorithm from random forest to LightGBM to improve model performance and scalability. Increase dynamic covariates (e.g., antecedent precipitation) and optimise feature selection routines (Bayesian optimisation).
- **Implement a finer spatial resolution** by incorporating the (newly available) 30 m GPP product (this increases computational complexity ca. 266-fold, it might be necessary to step up to more scalable compute environment (e.g., AWS Compute) and/or reduce study area down from the current regional-scale).
- Undertake in **deeper evaluations to ensure reliability of implementation and interpretations of the refined RPI implementation**, including:
    - Examination of attributes of the pixels with the worst temporal performance, with 2-dimensional evaluation of temporal performance against (possible/used) covariates).
    - Exploring spatial structure in covariates versus model performance (semivariograms)
- **Supporting AWF in their prioritisation of areas for landscape-scale conservation, informing resource allocation.** Especially to support greater objectivity in selection of 'priority restoration areas' as long as community willingness etc. is given greater weight.
- **Identification of degradation 'hotspots' as areas of elevated concern due to low mean RPI and/or negative RPI trends for investigation** using either on the ground and/or fine-resolution optical imagery (this might highlight the spatial effects of bomas and/or piospheres around natural and/or artificial water points).
- **Careful evaluation against areas with known differences in governance models,** including between areas where AWF has undertaken different interventions, and between e.g. individual ranches within the coarser groupings evaluated in the foundational dissertation work to reveal more nuanced differences in relative performance between management units.
    - Assessing one-dimension of the impact of conservation interventions like the designation of protected areas, targeted livestock grazing plans, and/or changes in land management etc.
    - For example, potentially focussing on the challenges of managing grazing in Amboseli, considering the impact of fencing and the fluidity of livestock movement. The fenced Amboseli Land Owners Conservation Association (ALOCA) area and the livestock-free Kimana sanctuary could be evaluated as potential reference sites.

- Produce a clear **summary report on the management applications of the Relative Productivity Index (RPI),** including clarifying the value of RPI as a tool for adaptive management in rangeland landscapes and how RPI could be integrated into AWFs landscape management workflows both as a monitoring and evaluation indicator and to identify degradation hotspots for prioritisation in restoration areas